



Impact of Inflation on Major Financial Ratios

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Received: 23 Mar 2020

Revised: 26 Apr 2020

Accepted: 28 May 2020

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ABSTRACT

Price remains constant over a period of time and fluctuates due to various reasons. The price floatation cause two types of economic conditions, one is Inflation and another is Deflation. Inflation may well be demarcated as increase in the general price level involve in the influences of production whereas deflation causes fall in the general price level. This study investigates the impact of inflation accounting application on companies' financial performance and financial position by analyzing financial ratios. The financial statements of 42 manufacturing companies covering 7 industrial sectors have been restated in current purchasing power for a period of 5 years (2004-05 to 2008-09). The ratios were calculated on both historical and adjusted numbers of financial statements to form two sets of ratios. In this study the Current Purchasing Power method, financial ratio analysis and different statistical tools like descriptive statistics and t-test have been employed to study the impact of inflation on major financial ratios. This study derived valuable information and results showed that with the impact of inflation, there is significant difference in liquidity, profitability and activity ratios.

Keywords: Manufacturing Firms, Financial Ratios, Inflation, descriptive Statistics, t-test

INTRODUCTION

Inflation continues to be an important reality of economic life in almost all countries. High inflation rates have seriously wind-swept monetary standards in these countries over the past few decades, and have brought forth new patterns of economic behavior. Despite of that, the effects of inflation on the financial performance of economic units go unrecognized in published financial statements. The primary purpose of the financial statements of a company is to give a true and fair picture of financial performance (i.e. profit or loss) of the company for a particular period and financial position (assets and liabilities) of the company for a particular point of time. Those transactions, which can be measured in terms of money, are taken into consideration for recording in the books of account. Money

23024



**Pramod Kumar Patjoshi**

Measurement Concept is a basic characteristic of accounting.

As per this accounting concept, only those business transactions which are capable of being expressed in terms of money are open for recording by the accounting system. It is also assumed that monetary unit, recording the business transactions, is stable in nature. However, price does not remain constant over a period of time. They tend to change due to various economic, social, political factors. Changes in the price level cause two types of economic conditions, one is Inflation and the other is Deflation. Inflation may be defined as period of general increase in the price level, whereas deflation causes fall in it. These changes in the price level lead to inaccurate presentation of financial statements which otherwise are prepared to present true and fair view of the company's financial health. Another point to consider for preparing the financial statements is the historical cost. In India the financial statements used to present according to actual and cost concept. But this Historical Cost Concept is based on the assumption that the unit of account, i.e. rupee has static value. Moreover, the purchasing power used to change by the passage of time and in India the purchasing power of the rupee is changing and it's rising continuously.

THE RESEARCH PROBLEM

In the past few years of high inflation, companies have reported very high profits on the one hand but on the other, they have faced real financial difficulties. This is so because in reality dividend and taxes have been paid out of capital due to overstated profits arrived by adopting the historical cost concept. This overstatement of profits can have several further effects. When these reported profits are used as the basis of corporate taxes to the Government, it leads to larger tax payments, and the Government becomes an important beneficiary of inflation. Furthermore, if reported profits are used as a basis of corporate decision making, the companies may not be setting prices sufficiently high to ensure an adequate rate of return on a long-term basis, if they have some scope for pricing policies. In addition, dividends are paid out to an excessive degree and thereby they have inadequate level of internal reserves to maintain their real resources intact.

The effect of inflation on the interpretative value of financial statements is much pronounced and it is more frequently put forward as an argument in favour of devaluation from the existing historical cost accounting' system. There are two important aspects of this problem worth monitoring. Firstly, during the period of prolonged inflation various items of the balance sheet, based on different levels of costs and prices, are not comparable in any real sense. In the Profits and Loss account, inventory profits and capital gains get inextricably mixed with operating profit thus making the proper assessment of the earning capacity of the firm difficult, if not impossible. In nutshell, the financial statements become difficult to interpret and their use as a tool of managerial decision-making is much lessened. Secondly, such financial statements mislead the shareholders and other users. The concept of 'profit' and 'maintenance of capital' based on the monetary postulates make the shareholders believe that so long as their money capital is maintained, their interest in the company are fully protected. This belief is, however, erroneous and quite misleading. The real interest of shareholders lies in the yield from business as a going concern and in the eventuality of its winding up in its actual break-up value as opposed to apparent book value. As a consequence of these interpretative difficulties, shareholders and other investors are not provided with information, which enables them to interpret the operating results and to judge the relative effect of price level changes upon a particular enterprise. Such inability arising due to interpretative difficulties of historical cost accounting to properly assess the business position may result in lost business opportunities.

Traditionally, historical cost based accounting information about the operations of companies has ignored the effects of inflation. But the users of financial information, such as current and potential investors, creditors, lenders, suppliers, customers, employees, government authorities and public, need relevant and reliable information about the financial position, performance and changes in the financial position of firms for making economic decisions. Inflation, on the other hand, distorts financial information by creating an impact on the firm's operational and financial results. In a hyperinflationary economy, reporting of operating results and financial position without



**Pramod Kumar Patjoshi**

restatement is misleading and thus is not useful (International Accounting Standards 29, code 2). Therefore, it is necessary that financial statements reflect the true picture and are free from the negative effects of inflation. Even though inflation accounting has long been debated, a necessary attention is avoided to provide its effects on financial statements of businesses. Hence, this study is conducted to present empirical evidence on the results of inflation on financial statements through analysis of some major ratios. For the purpose, 10 ratios categorized under three groups are analyzed.

LITERATURE REVIEW

Gupta Ramesh and Bhandari L C (1978), in their working paper mention that the whether accountants should be required to adjust reported income for inflation. The objective of this article is to measure the impact of inflation on reported profits and relevant financial ratios. The earnings of 57 companies covering 9 industries have been restated for a period of 7 years (1970-1976). The results emphasize the differential effects on companies with varying inflation rates with general price level adjustments and the significance of monetary gains and losses. The effects of restatement on dividend coverage and tax burden have been suitably highlighted. D. J. Daly (1982), in the article "Inflation, Inflation Accounting and its Effect, Canadian Manufacturing, 1966-82", provides estimates of the effects of inflation in Canada on the reported rate of return in manufacturing firms from 1966 to 1982. It provides estimates for several different concepts of rate of return (both for all assets, whether financed by equity or debt, and for the narrower equity to the owners) and for both a narrow and wide range of financial assets. Comparisons are made with similar studies for the United Kingdom.

Shalom Hochman and Oded Palmon (1985) in their article "The Impact of inflation on the Aggregate Debt-Asset Ratio" demonstrate the impact of inflation on the aggregate debt-asset ratio cannot be determined theoretically. However, it is shown that inflation is likely to increase this ratio when personal income tax schedules are indexed to the price level and/or when leverage-related costs are relatively high and the personal tax rate on income from holding common stocks is relatively low. Whittington G., Saporta V and Ajit Singh (1997), in their working paper "The Effects of Hyper-Inflation on Accounting Ratios Financing Corporate Growth in Industrial Economies" described the hyper-inflation can have a severe distortionary effect of the pattern of corporate finance which is apparent from company accounts. A simple algorithm, based upon the method of inflation accounting applied in Brazil, is developed and applied to the accounts of Turkish listed companies for the period 1982-90. The adjusted figures give a more plausible picture of corporate profitability and growth, and this suggests that the adjustment method is substantially successful.

Ambrish Gupta (2000), in his research entitled to "Inflation Accounting- The Indian Context", this study was a modest effort towards a systematic and comprehensive analysis of various aspects for inflation accounting and looks for offering an acceptable solution to this problem in the Indian context. It also made an assessment of the its effect of inflation on the profitability plus financial position, respectively, of the corporate entities, in addition to above it attempt to make an overall review of the financial statements, through ratio analysis and funds flow analysis, in the light of inflation. This study moreover reflects effects of inflation, over sixteen years between 1983-84 to 1998-99, on the financial health of Oil India Ltd. Karapinar A. and Zaif F., (2005), in their article "Enflasyon Muhasebesinin Finansal Tablolar Analizine Etkisi, (The Effect of Inflation Accounting on Financial Statement Analysis)" In their study, Karapinar and Zaif examined the effects of inflation on accounting practice of companies' financial ratios. Their sample covered the 73 non-financial companies listed Istanbul Stock Exchange as of 2003. The ratios were calculated on both historical and adjusted numbers of financial statements to form two sets of ratios. Results showed that there was no significant change in liquidity, financial, profitability and activity ratios except fixed asset turnover ratios. Akdoğan, Aktas and Unal, in their study in 2009, extended the number of companies in the sample of Karapinar and Zaif. The results covering 146 companies were consistent with the findings of Karapinar and Zaif's study. Their results revealed that a statistically significant change for the whole sample occurs only on Total Assets Turnover. Other ratios did not show any considerable difference.



**Pramod Kumar Patjoshi**

Charles N'cho-Oguee, Daniel L. Blakley, L. William Murray, and Marolee Beaumont Smith (2011), in their article "Econometric Analysis of Functional Relationship between Inflation and Growth of Firms in South Africa: Empirical Research Findings" this research is to investigate the impact inflation and other factors on the growth of business firms operating in South Africa. Data sets of South African firms' financial statements over the period of 1983-1990 were assembled to permit a detailed examination of the impact of inflation on firm's financial ratios. It has concluded that firm's debt-to-equity, sales-to-assets, and profitability ratios are all positively associated with growth and adversely affected by high inflation; a firm's working capital-to-sales ratio is negatively related to growth and is positively affected by high inflation; and there is a real, measurable impact of the financial instabilities associated with apartheid on firm's growth. Aydın Karapinar, Figen Zaif and Rıdvan Bayirli (2012), this study investigates the impact of inflation accounting application on key financial ratios. These studies related to the financial statements of 132 companies listed in the Istanbul Stock Exchange (ISE) are studied. An analysis of paired samples t test has been conducted on the financial ratios of the companies. The results showed that a significant difference between adjusted cost based financial ratios and historical cost based financial ratios occurs only for current, ratios, equity ratios and noncurrent turnover ratios. The study offered valuable information as to analyzing companies operating in hyperinflation economies. In India serious thinking on having to adjust historical cost accounts to price level change has been rather few and far between.

Patjoshi (2013), in the study investigated the comparison between reported and inflated financial ratios. Financial statements of 42 manufacturing companies covering 7 industrial sectors have been restated in current purchasing power for a period of 5 years (2004-05 to 2008-09). The ratios were calculated on both historical and adjusted financial statements to form two sets of ratios. Similarly, by considering the above data Patjoshi (2013) studied impact of inflation on reported financial performance: of Indian corporate sector, Patjoshi, (2013) examined the effect of inflation on measures of financial performance on Indian manufacturing industry, Patjoshi (2013) scrutinized impact of inflation on gross value added and net value added, Patjoshi (2014) observed impact of inflation on corporate profitability and also Patjoshi (2016) observed impact of inflation on major stakeholders. All the above studies used the Current Purchasing Power method and different tools of financial statement analysis like comparative and common-size statement analysis.

OBJECTIVES OF THE STUDY

The objectives of the proposed study are to find the impact of inflation on financial performance & position through analyzing the major financial ratios. The objectives of the proposed study are

- a) To study the impact of inflation on short term solvency of sample companies through liquidity ratios.
- b) To analysis the impact of inflation on financial performance of sample companies through profitability ratios.
- c) To find out the impact of inflation on efficiency of resources employed by the sample companies through activity ratios.

HYPOTHESES FOR THE STUDY

To study the research problems and to attain the research objectives, three hypotheses have framed. Broadly, we have attempted to test the null hypothesis against the alternative hypothesis. The null hypothesis and the alternative hypothesis framed for the purpose are:

1. There is no significant difference in between reported and inflated liquidity ratios.
2. There is no significant difference in between reported and inflated profitability ratios.
3. There is no significant difference in between reported and inflated activity ratios.



**Pramod Kumar Patjoshi**

METHODOLOGY AND TESTS USED IN THE STUDY

The work conducted is a study of 42 undertakings, selected randomly from manufacturing sectors operating in India. The companies so selected are capital intensive, where there is a heavy investment in fixed assets and inventories, profitable and following the same accounting practices throughout the period of study. These sample companies belong to different sectors, viz. Auto, Cement, Chemical, Fertilizer, Food, Petroleum and Steel.

The year-end financial statements of sample companies were used for the comparing the reported and inflated performances. The published annual reports, books, journals, web pages, etc. of the selected companies form the main sources of information. The data so collected are analyzed with the help Current Purchasing Power Method (CPP), Financial Statement Analysis (FSA) and Statistical tools such as; Average, Variance Standard Deviation, Kurtosis, Skewness, and t-test are employed too to draw meaningful conclusion. The t-test is used to compare the values of the means from two groups. The two sample of t-test has been performed because the variances of two groups are assumed to be unequal.

Current Purchasing Power Method

Current Purchasing Power Method of accounting requires the companies to maintain the financial statements on conventional historical cost basis, but it further requires presentation of supplementary statements in items of current purchasing power of currency at the end of the accounting period. In this method the various items of financial statements, i.e. balance sheet and profit and loss account are adjusted with the help of recognized general price index. The consumer price index or the wholesale price index prepared by the Reserve Bank of India can be taken for conversion of historical costs. However, WPI (All Commodities) is being used in this study,

Conversion Process

For analyzing the impact inflation on financial performance the Historical Cost Based (HCB) accounting, financial statements for all the years from 2004-05 to 2008-09 were converted into Accounting for Current Purchasing Power (CPP) financial statements in terms of the index number prevailing in the month of March 2009. The adjustments for inflation are based on movements in wholesale price index has presented in Table-1.

Source: Handbook of Statistics on Indian Economics: RBI, 2008-09 Sept15 2009, Office of Economic Advisor Ministry of Commerce and Industry.

The conversion process is explained hereunder

- (a) All items of Profit & Loss Account, except Inventory Cost, Depreciation, Taxation, and Equity Dividend have been restated with reference to the "average price index of the year/period" as applicable to the individual year.
- (b) Inventory cost has been restated after segregating opening balance of inventories, purchases of raw materials and closing balance of inventories as follows:
 - Opening balance of inventories restated in previous year average price index.
 - Closing inventories and purchases of raw materials restated in average year price index as applicable to the individual year.
- (c) Fixed Assets and Depreciation cost of all the years of study has been adjusted to year base year 2000-01 at year end price index.
- (d) Taxation, Dividend on equity shares have been restated with reference to the "end of the year/period index" as applicable to the individual year
- (e) The CPP Method divides the Balance Sheet items into two categories: Monetary items and Non-monetary items. Monetary items are those assets and liabilities the amounts of which are fixed by contract or statute in



**Pramod Kumar Patjoshi**

terms of the number of rupees irrespective of the changes in the purchasing power of rupee. Items which comes under monetary in nature are as follows:

- *Monetary assets* include Investments, which are fixed in rupees, Current Assets other than Inventories.
- *Monetary Liabilities* include Secured Loans, Unsecured Loans, Current Liabilities and Provisions Since the value of monetary items is fixed in rupees, they are already expressed in terms of current purchasing power of rupee and, therefore, need no restatement.

For Calculating purchasing power gain/loss, the balance of net monetary liabilities/assets as on the date of the Balance Sheet is bifurcated into opening balance and additions/decrements thereto during the year. The opening balance is restated with reference to the index prevalent on that date. Additions/decrements are restated with reference to the average index of the year. The closing balance is deducted from the total of restated opening balance and additions/decrements. The resultant figure, if positive, is gain otherwise loss in the case of net monetary liabilities and vice versa in the case of net monetary assets.

After converting the Historical Based financial statements into Current Purchasing Power, the major financial ratio has been calculated. The calculated ratios are presented in Table-2.

DATA ANALYSIS AND DISCUSSION**IMPACT OF INFLATION ON LIQUIDITY RATIOS**

These ratios are calculated to comment upon the short-term paying capacity of a firm or a concern's ability to meet its current obligation. The important liquidity ratios are current ratio and quick ratio.

Statistical Outcome of Liquidity Ratios

Table-3 summarizes the results of Liquidity Ratios under HCB method as well as CPP method from 2004-05 to 2008-09 by the help of descriptive statistics and t-test.

Descriptive statistics and t-test for the current ratio provide that mean of reported current ratio is less as compared to that of the inflated; leading to the conclusion that liquidity position of sample companies is better under CPP method. On the contrary, lower standard deviation for reported current ratio as compared to inflated current ratio clearly indicates that former is more consistent than the latter. Even though value of kurtosis is found to be more than 3 under both the methods, it is higher in CPP as compared to HCB method. Therefore it can be concluded that the inflated current ratio is more peaked than that of the reported current ratio. Yet again, the correlation value is 0.9998 represents high degree of positive correlation between both the methods. The p-value of 0.0188, which is less than 0.05, indicates a significant difference in the value of current ratio between HCB and CPP methods at 5 percent level of significance.

Hence in the case of Current Ratio our hypothesis (H_{03}) is rejected as there is a significant difference between two accounting methods.. The Descriptive statistics and t-test analysis for the quick ratio in Table-3 reveals that the mean, standard deviation, kurtosis, skewness and all other findings are the same under both the accounting methods (HCB and CPP). It is because inventory is taken away from current assets for finding out the quick assets, the result indicates the reported and inflated quick ratios are identical.



**Pramod Kumar Patjoshi****IMPACT OF INFLATION ON PROFITABILITY RATIOS**

This group of ratios measures the overall performance and effectiveness of the firm. The main profitability ratios include Gross Profit Margin, Operating Profit Margin, Net Profit Margin, Return on Investment and Dividend Payout Ratio etc.

Statistical Outcome of Profitability Ratios

Generally, profitability ratios measure the financial efficiency of firms in different ways. The descriptive statistics as well as results of t-test of profitability ratios viz. gross profit margin, operating profit margin, net profit margin and return on investment both under HCB and CPP methods from 2004-05 to 2008-09 is given below in Table -4.

The Table-4 clearly shows that mean value of gross profit margin is higher in the reported (HCB) method as compared to inflated (CPP) method for sample companies over the period of study (from 2004-05 to 2008-09), meaning thereby reported gross profit margin has performed better than inflated gross profit margin. But the HCB gross profit margin has more variation than that of the CPP, since both standard deviation as well as variance shows higher value in HCB method than CPP method. Though, the distribution of gross profit margin are found to be positively skewed in both the methods, the CPP gross profit margin is little bit higher skewed than that of the HCB gross profit margin. These ratios are observed to be platykurtic by nature i.e. it is more flat on HCB method. The p-value of 0.0024 in case of gross profit margin indicates significant difference in the absolute value of gross profit margin between reported and inflated at 5% degree of significance.

In case of operating profit margin, the mean value under CPP method is less than the HCB method, disclosing thereby operating profit margin has reported at a higher value in comparison to actual operating profit margin adjusted to inflation. The standard deviation of reported operating profit margin is less than inflated which reveals that there is less variation in reported operating profit margin in comparison to inflated. Skewness of operating profit margin found to be positive in case of HCB method, where as it is negative in case of CPP method. So HCB operating profit margin is positively skewed and for CPP it is negatively skewed. The p-value shows a significant difference between reported and inflated with respect to operating profit margin. The factors influencing the inflation were tested for the degree of relationship among them to find whether the fluctuation in one factor affects the other factors. To identify the same, the factors were measured for bi-variate correlation with respect to each other and their coefficients were given in the Table-4. From the Table-4, it is clear that there is a significant correlation existing between the factors of operating profit margin of reported and inflated.

Similarly, the mean of net profit margin is higher in case of HCB method as compared to CPP method. The variation of reported net profit margin is less than the inflated net profit margin as both standard deviation and variance is found to be less in HCB method. Here too there is a significant difference between reported and inflated net profit margin as the p-value is less than 0.05. However the skewness under both HCB and CPP method was found to be negative. So it can be said that net profit margins are negatively skewed. Net profit margins under both the methods experimented to be leptokurtic by nature i.e. they are peaked.

From the Table-4, it is evidenced that Return on Investment (ROI) has executed the same trend like that of gross profit margin, operating profit margin and net profit margin. The mean value and the standard deviation of Return on Investment (ROI) under HCB method is found higher than that of CPP method. Therefore there is less consistency in case of reported return on investment. The value of p shows that there is a significant difference between reported and inflated return on investment among the two accounting methods (HCB and CPP) for the study period. While in the case of skewness both reported and inflated return on investment are positively skewed and HCB return on investment is higher than the CPP return on investment. Both the reported and inflated returns on investment are leptokurtic by nature. The HCB return on investment is more peaked than the return on investment of CPP.



**Pramod Kumar Patjoshi**

Table 4 shows the results of the profitability ratios, where it can be observed that the entire profitability ratios have dropped significantly. This shows that inflation adjustment leads to impact upon the shareholders' funds. The 3rd hypothesis does suggest no significant difference between the ratios of two groups (HCB and CPP). But the p-values of 0.0024, 0.0003, 0.0000 and 0.0006 respectively for gross profit margin, operating profit margin, net profit margin and return on investment shows that there is significant difference between reported and inflated profitability ratios. Therefore hypothesis (H_{03}): 'there is no significant difference in reported and inflated financial ratios' is rejected.

IMPACT OF INFLATION ON ACTIVITY RATIOS

Activity ratios are calculated to measure the efficiency with which the resources of a firm have been employed. These ratios are also called turnover ratios because they indicate the speed at which the assets are being turned over into sales.

Statistical Outcome of Activity Ratios

Activity ratio is a very important tool to measure the velocity of current assets of an organization and includes generally Creditor Turnover Ratio, Debtor Turnover Ratio and Inventory Turnover Ratio. The details of descriptive statistics and t-test of these ratios for sample companies under HCB method as well as CPP method are given away in Table -5 for the period under study.

It is observed from the Table -5 that the creditor turnover ratio has performed better in the case of CPP method as compared to HCB method, which indicates creditors are treated well in case of inflationary condition. The standard deviation as a measure of variation is found to be higher in CPP creditor turnover ratio than that of the HCB. Again, the creditor turnover ratio under both the methods is found to be positively skewed, but it is more skewed under CPP method than the HCB method. The ratio is experiential to be platykurtic by nature i.e. it is more flat in case of HCB method. As the p-value comes to 0.0000, so it is concluded that there is significant difference in creditor turnover ratio under both the accounting methods under discussion. Similarly, it is observed from the Table-5 that the mean value of debtor turnover ratio is superior under the CPP method as compared to that of HCB method but having more variations (as the standard deviation is found to be higher). Thus it can be said that inflation has favorable impact debtors of the sample companies during the period of study. Yet again the ratio is found to be positively skewed and leptokurtic under both the methods of accounting, but more skewed as well as more peaked under the HCB method than CPP. Like the creditor turnover ratio, there is significant difference in values of this ratio under both the accounting methods since value of p is found to zero.

On the contrary, The mean value, standard deviation as well as variance of inventory turnover ratio is found to be higher in HCB method as compared to the CPP method implying that the production efficiency of sample companies has suffered to greater extent due to inflation, but it is inconsistent during the period of study. Here also, like other turnover ratios, we found the inventory turnover ratio is positively skewed under both the methods and marginally more peaked under HCB method. There is also significant difference in values of inventory turnover ratio under both the accounting methods as evidenced from the p-value.

As the p-values for all three activity ratios (Creditor Turnover Ratio, Debtor Turnover Ratio and Inventory Turnover Ratio) are found to be equal (0.0000), there is a significant difference in between two accounting methods. For this reason hypothesis (H_{03}) for Activity Ratios is discarded.

CONCLUSION

From the above discussion and interpretation of the descriptive statistics and t-test of key financial ratios, undoubtedly it is observed that inflation has affected all the ratios under study except the quick ratio. With the





Pramod Kumar Patjoshi

pressure of inflation the current ratio, creditor turnover ratio and debtor turnover ratio have changed and performed better, but no change has occurred to the quick ratio. Nevertheless all the profitability ratios and inventory turnover ratio have suffered badly due to the impact of inflation. The financial ratio analysis thus confirms that historical accounts overstate profitability and understate liquidity. From the above discussion, it is clear that due to inflation the profitability of sample companies has been understated, resulting additional payment of tax and dividend, but reduction in shareholders' funds by dropping retained earnings. Moreover, the dismal performance revealed under the CPP method points towards the efficiency of the management in fighting against inflation. Whatever view one takes, no one can deny that, but inflation has taken its toll. The reason for overstatement of corporate profitability under HCB method may be attributed to undervaluation of material cost and depreciation. The performance which otherwise appears to be quite good, turned out to be very dismal, when adjustments for current purchasing power of rupee are made. The CPP method adjustments have thus proved that historical profitability is a fairy story.

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Pramod Kumar Patjoshi

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Table No. – 1 Wholesale Price Index in India [2000-09]

Year	Average	Average as per 2004-05	Year End	Year End as per 2004-05
2000-01	83.19	100.00	84.00	100.00
2001-02	86.18	103.59	85.48	101.76
2002-03	89.12	107.13	90.60	107.86
2003-04	93.98	112.97	94.93	113.01
2004-05	100.07	120.29	100.00	119.05
2005-06	104.50	125.62	105.70	125.83
2006-07	111.40	133.91	112.80	134.29
2007-08	116.60	140.16	121.50	144.64
2008-09	126.00	151.46	123.50	147.02

Table No. – 2 Ratios Used in the Study

Liquidly Ratios	Profitability Ratio
Current Ratio	Gross Profit Margin
Quick Ratio	Operating Profit Margin
Activity Turnover Ratio	Net Profit Margin
Debtor Turnover Ratio	Return on Investment
Creditor Turnover Ratio	Dividend Payout Ratio
Inventory Turnover Ratio	

Table No. – 3 Statistical Results of Liquidity Ratios

Particulars	Current Ratio		Quick Ratio	
	HCB Method	CPP Method	HCB Method	CPP Method
Mean	2.1681	2.2620	1.2038	1.2038
Standard Deviation	3.0700	3.3454	0.7239	0.7239
Sample Variance	9.4249	11.1917	0.5241	0.5241





Pramod Kumar Patjoshi

Kurtosis	34.5451	35.3145	1.2978	1.2978
Skewness	5.6496	5.7361	1.2626	1.2626
Pearson Correlation	0.9998		1.0000	
Hypothesized Mean Difference	0.0000		0.0000	
t Stat	-2.1502			
P(T<=t) one-tail	0.0188			
t Critical one-tail	1.6839			
P(T<=t) two-tail	0.0376			
t Critical two-tail	2.0211			

Table No.-4 Statistical Results of Profitability Ratio

Particulars	Gross Profit Margin		Operating Profit Margin		Net Profit Margin		Return on Investment	
	HCB	CPP	HCB	CPP	HCB	CPP	HCB	CPP
Mean	0.3348	0.3232	0.1069	-0.0617	0.0618	0.0032	0.4169	0.1551
St. Deviation	0.2616	0.2606	0.1231	0.3359	0.1030	0.1581	1.3087	0.3215
Sample Variance	0.0684	0.0679	0.0152	0.1128	0.0106	0.0250	1.7127	0.1034
Kurtosis	-0.8664	-0.8272	2.4558	2.4587	8.4900	18.0787	40.1024	14.1382
Skewness	0.4850	0.5277	0.0393	-1.7679	-1.8331	-3.5369	6.2678	3.1014
Pearson Correlation	0.9953		0.5010		0.9212		0.5676	
Hypothesized Mean Difference	0.0000		0.0000		0.0000		0.0000	
T Stat	2.9901		3.7177		5.1103		3.4662	
P(T<=t) one-tail	0.0024		0.0003		0.0000		0.0006	
T Critical one-tail	1.6839		1.6839		1.6839		1.6839	
P(T<=t) two-tail	0.0048		0.0006		0.0000		0.0013	
T Critical two-tail	2.0211		2.0211		2.0211		2.0211	

Table No. -5 Statistical Results of Activity Ratios

Particulars	Creditor Turnover Ratio		Debtor Turnover Ratio		Inventory Turnover Ratio	
	HCB Method	CPP Method	HCB Method	CPP Method	HCB Method	CPP Method
Mean	3.7426	4.0907	22.9793	25.1120	11.4766	11.1810
Standard Deviation	2.8469	3.1267	23.7664	25.9161	7.8027	7.5617
Sample Variance	8.1050	9.7761	564.8427	671.6457	60.8817	57.1790
Kurtosis	-0.1930	-0.0783	5.3379	5.2297	3.1295	3.0977
Skewness	0.7862	0.8134	2.2205	2.2088	1.5887	1.5714
Pearson Correlation	0.9997		0.9998		0.9994	
Hypothesized Mean Difference	0.0000		0.0000		0.0000	
t Stat	-7.5652		-6.0256		5.5831	





Pramod Kumar Patjoshi

P(T<=t) one-tail	0.0000		0.0000		0.0000	
t Critical one-tail	1.6839		1.6839		1.6839	
P(T<=t) two-tail	0.0000		0.0000		0.0000	
t Critical two-tail	2.0211		2.0211		2.0211	

